

Index sheet for Course file

Sl. No.	Contents	Page No.
1	Academic calendar of the University	2
2	Course outline (Syllabus of the course) (template – 1)	3
3	Lesson plan (template – 2)	5
4	Timetable (of all sections of that course)	13
5	Student list (template – 3) (extracted from SAP)	15
6	Continuous Internal Evaluation (CIE)	
	- Question paper for the CIE components (min – n +1; max – n + 4) To be common across all sections	20
	- Scheme of evaluation (common) and rubrics of all components	23
	- 2 sample answer scripts/assignments of the students (of each CIE component) (one best script, one average script)	
	- Consolidated marks sheet of CIE (of all components)	27
	- List of slow learners (students whose score is less than 40% in a CIE will be termed as slow learners)	31
	Remedial class details - Time table of the class to be conducted - Classes work done statement - Retest (Question paper and answer sheets) (which includes the details of the classes taken, retest conducted and the improved marks from the retest to be considered for CIE)	
	- Finalized CIE marks signed by all the students (which is submitted to the exam office)	
7	- SA & FI list	
	- Final attendance sheet (mentioning the % of class attended)	
8	Semester End Examination (SEE)	
	- Question paper	
	- Scheme of evaluation	
	- Evaluation rubrics	

	- Result sheet of CIE	
9	Course-end survey (questions to be specific to the course; not more than 8)	
10	Course attainment	

Course outline (Syllabus of the course) - Template – 1

Course Code: CS3508 Name of the Programme: BTech (H) Name of the Course: Cloud Security and Migration Semester: 6th sem		
Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
3	2+0+2	30:0:30=60
Course Objectives:		
Explain fundamental cloud security principles, architectures, service models (IaaS, PaaS, SaaS), and the shared responsibility model to effectively manage risks in cloud environments.		
Apply data protection strategies (e.g., encryption, data loss prevention) and implement identity and access management (IAM) mechanisms to enforce secure access control in cloud platforms like AWS and GCP.		
Analyze network security controls (e.g., firewalls, VPCs), vulnerability assessment techniques, and incident response frameworks to secure and respond to threats in cloud infrastructure.		
Design and plan cloud migration strategies using industry-standard frameworks (e.g., AWS Well-Architected, Google Cloud Adoption Framework) and tools, while evaluating security best practices for compliance, governance, cost optimization, performance, and continuous monitoring.		

Syllabus:

Module – 1	No. of hours
Cloud Security Foundations	6
Security Concepts: Principle of Least Privilege, Defense in Depth, Zero Trust, Threat Actors and Trust Boundaries; Cloud Architecture & Models: Cloud Service Models: IaaS, PaaS, SaaS, Deployment Models: Public, Private, Hybrid, Shared Responsibility Model; Risk Management: Cloud Risk Assessment, Regulatory and Compliance Overview	

Module – 2	No. of hours
Data, Identity & Access Security	6
Data Security: Data Classification & Protection, Encryption Techniques, Tokenization, Compliance: GDPR, HIPAA; Identity & Access Management (IAM): IAM Lifecycle: Request, Approve, Grant, Revoke, Authentication Mechanisms: MFA, API Keys, Federated Identity & Single Sign-On (SSO), Secrets Management	

Module – 3	No. of hours
Network Security, Vulnerability & Incident Response	6
Network security: Firewalls, DMZ, NAT, VPC, IPv6, Zero Trust Networking, SDN & NFV, Data Loss Prevention (DLP), Network Segmentation; Vulnerability Management: Scanning Tools (SAST, DAST, IAST), Manual Testing and Penetration Testing, Patch & Configuration Management; Incident Response: Threat Detection, SIEM, MITRE ATT&CK, OODA Loop, Forensics, Recovery & Post-Incident Analysis	
Module – 4	No. of hours
Cloud Migration Strategies and Practices	6
Migration Fundamentals: Why Migrate to Cloud?, Migration Challenges and Considerations; Migration Planning: Assessing Existing Infrastructure, Cloud Readiness and Choosing a Model; Migration Strategies (6 Rs): Rehost, Replatform, Repurchase, Refactor, Retain, Retire; Migration Execution: Migration Phases: Discover → Plan → Migrate → Operate, Migration Tools (AWS Migrate, Azure Migrate, GCP Migrate), Post-Migration Performance Tuning	

Module – 5	No. of hours
Secure Cloud Migration	6
Security Challenges During Migration: Data Exposure & Transit Risk, Legacy System Threats; Secure Migration Practices: Data Encryption (At Rest & In Transit), VPN, Dedicated Connections, IAM Hardening; Compliance and Governance: Real-Time Logging, Audit Trails, Monitoring, Cloud-Native Security Integration; DevSecOps & Continuous Security: Integrating Security into CI/CD Pipelines, Secure Multi-cloud and Hybrid Environments; Case Studies: Real-Life Secure Cloud Migrations, Industry Best Practices	

Course outcomes
CO1: Cloud Security Fundamentals Understand fundamental cloud security concepts, including service models (IaaS, PaaS, SaaS), deployment models, security architecture, and the shared responsibility model.
CO2: IAM, Data Protection, and Compliance Apply identity and access management (IAM) mechanisms, data protection strategies (e.g., encryption), and compliance enforcement techniques in cloud environments.
CO3: Network Security and Incident Response Design and implement network security configurations, conduct vulnerability assessments, and develop incident response procedures for cloud-based systems.
CO4: Secure Cloud Migration Planning Plan and execute cloud migrations using industry best practices and tools, integrating security across the lifecycle with DevSecOps, governance, monitoring, performance, cost, and

reliability considerations.

Text Books

1	Chris Dotson: Practical Cloud Security (2023), O'Reilly, ISBN: 9781098148171
2	John R. Vacca (Editor): Cloud Computing Security Foundations and Challenges (2017), Taylor & Francis Group, ISBN: 9781482260946
3	Steve Swoyer: Migrating Applications to the Cloud (2021), O'Reilly, ISBN: 9781098102791
4	Michael C (McDonald Bradley Inc Sierra Vista Arizona) Daconta: Great Cloud Migration (2013), Outskirts Press, ISBN: 147872255X

Reference books

1	Ben Malisow: (ISC)2 CCSP Certified Cloud Security Professional Official Study Guide (3rd Edition) (2020), ISBN: 9781119603375
2	Josh Armitage: Cloud Native Security Cookbook Recipes for a Secure Cloud (2022), O'Reilly, ISBN: 9781098106300
3	Kieran Broadfoot: A Practical Guide to Cloud Migration (2021), O'Reilly, ISBN: 9781492095170

Lab Practicals
Experiment 1: Identity & Access Management (IAM) Foundations – create users, roles, and a Permissions Boundary mock-up to illustrate shared-responsibility.
Experiment 2: Build & Test a “Cloud Firewall” Using Security Groups – harden a two-tier VPC, scan ports from an EC2 bastion, and verify SG ingress/egress rules.
Experiment 3: Vulnerability Assessment with Amazon Inspector – run Inspector 2 scans on an EC2 instance.
Experiment 4: Client-Side Encryption with AWS KMS & the Encryption SDK – encrypt data locally, upload to S3, and decrypt on download
Experiment 5: AWS Application Migration Service (MGN) – Migrating an existing server or application to AWS with minimal changes using a rehosting strategy.
Experiment 6: AWS Database Migration Service (DMS) – Migrating relational databases to AWS with minimal downtime.
Experiment 7: Migration of Object Storage Data Between Amazon S3 Buckets Using AWS DataSync.
Experiment 8: Continuous Compliance Logging with AWS CloudTrail Lake – create an event data store, write Lake queries for IAM changes & S3 policy updates, and set up SNS alerts.
Experiment 9: Migrate an EC2-based Web Application from Source AWS Account to Target AWS Account.
Experiment 10: GuardDuty detects the behavior and automatically isolates the instance from the network.

Lesson plan (template – 2)

Name of the Programme:	B.Tech (Hons)	
Title of the Course:	Cloud Security and Migration	
Course code	CS3508	
Semester:	VI	
Names of the Course Instructor(s)	Prof Aparna R Prof Thiruselvan	
Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
03	(2 + 0 + 2)	60

Ses sion	Modul e No.	Topic	Pedagogy /activities	Pre-reading/reference
1	1	Introduction to security	Presentation using online web resources	
2		Security Concepts (Principle of Least Privilege, Defense in Depth, Zero Trust, Threat Actors and Trust Boundaries)	Lecture with Interactive demo - configure least privilege IAM policy simulation	<i>Practical Cloud Security</i> Ch. 1;
3		Cloud Service Models (IaaS, PaaS, SaaS - definitions, examples, pros/cons)	Lecture/slides Poll: "Best model for a startup?" Debrief	AWS/GCP service model overviews; <i>Cloud Computing Security</i> Ch. 2
4		Deployment Models & Shared Responsibility (Public, Private, Hybrid; Shared Responsibility Model)	Lecture with diagrams (public=shared infra, hybrid=best of both) Case study discussion for all Deployment models	Syllabus Shared Responsibility section; AWS model infographic
5		Risk Management Introduction (Cloud Risk Assessment - identification, prioritization)	Lecture on risk frameworks (e.g., NIST) Risk assessment activity	<i>Practical Cloud Security</i> Ch. 2 risk assessment; Syllabus risk overview

6		Regulatory & Compliance Overview (GDPR, HIPAA, cloud impacts)	Lecture/table of regs vs. cloud models Compliance mapping exercise for data classification	Syllabus compliance list; GDPR/HIPAA summaries (CSA guides)
7		Integration, Lab Preview	Review full module mind map Hands-on mini-lab: Simulate Zero Trust boundary in diagram tool	Full Module 1 syllabus; Lab Exp 1 description
8 & 9		Experiment 1: Identity & Access Management (IAM) Foundations – create users, roles, and a Permissions Boundary mock-up to illustrate shared-responsibility		Lab manual Google drive link
10 & 11		Experiment 2: Build & Test a “Cloud Firewall” Using Security Groups – harden a two-tier VPC, scan ports from an EC2 bastion, and verify SG ingress/egress rules.		Lab manual Google drive link
12	2	Data Security: Classification & Protection	Lecture on data classification levels (public, internal, confidential, restricted) -Group activity - classify sample datasets and map protection controls	<i>Practical Cloud Security</i> Ch. 3;
13		Encryption Techniques	Lecture/demo of symmetric/asymmetric encryption, KMS Poll: "When to use encryption at-rest vs. in-transit?" Debrief	Textbook Ch. 3 Encryption Techniques;
14		Tokenization & Data Protection Strategies	Lecture comparing tokenization vs. encryption use cases Case study - design data protection for multi-tenant SaaS app	<i>Cloud Computing Security</i> Ch. 4
15		IAM Lifecycle & Compliance (GDPR, HIPAA)	Lecture on IAM lifecycle flowchart + compliance mapping table	Syllabus IAM Lifecycle; AWS IAM Best Practices doc
16		Authentication Mechanisms (MFA, API Keys, SSO)	Lecture/demo of MFA flows, federated identity diagrams	<i>Practical Cloud Security</i> Ch. 4

17		Secrets Management & Module Integration	Lecture on secrets management (Parameter Store, Secrets Manager) Activity - simulate secrets rotation policy implementation	
18 & 19		Experiment 3: Vulnerability Assessment with Amazon Inspector – run Inspector 2 scans on an EC2 instance.		Lab manual Google drive link
20 & 21		Experiment 4: Client-Side Encryption with AWS KMS & the Encryption SDK – encrypt data locally, upload to S3, and decrypt on download		Lab manual Google drive link
22	3	Network Security Fundamentals (Firewalls, DMZ, NAT, VPC)	Lecture with VPC architecture diagram and firewall rule examples Group activity - "NAT vs. IPv6 addressing trade-offs in cloud"	<i>Practical Cloud Security</i> Ch. 5;
23		Advanced Networking (Zero Trust Networking, SDN/NFV, Network Segmentation)	Lecture/demo Zero Trust model vs. perimeter security Poll: "Zero Trust implementation challenges?" Debrief	
24		Data Loss Prevention (DLP) & Network Segmentation	Lecture on DLP policies and segmentation strategies Case study - implement DLP for S3 data exfiltration prevention	<i>Cloud Computing Security</i> Ch. 6
25		Vulnerability Management (SAST/DAST/IAST Scanning Tools)	- 10 min: Lecture comparing scanning methodologies with tool matrix - 30 min: Workshop - analyze Amazon Inspector scan results (Lab Exp 3) - 15 min: Group prioritization of vulnerabilities by CVSS score - 5 min: Break	Lab Exp 3 (Amazon Inspector); Syllabus Vulnerability ScanningSyllabus.pdf
26		Manual Testing, Pen Testing & Patch Management	- 15 min: Lecture on pen testing methodologies (OWASP, MITRE) - 25 min: Simulate pen test walkthrough - bastion host to VPC scanning - 15 min: Design patch management workflow for EC2 fleet - 5 min: Key concepts poll	Syllabus Manual Testing; Lab Exp 2 VPC hardeningSyllabus.pdf

27		Incident Response (SIEM, MITRE ATT&CK, OODA Loop, Forensics)	- 10 min: Lecture on OODA loop + MITRE ATT&CK cloud tactics - 30 min: Group simulation - SIEM alert triage and forensics response - 15 min: Formative quiz (scenario-based IR questions) - 5 min: Homework - create post-incident analysis template	Syllabus Incident Response; Lab Exp 10 (GuardDuty); <i>Practical Cloud Security</i> Ch. 7Syllabus.pdf
28 & 29		Experiment 8: Continuous Compliance Logging with AWS CloudTrail Lake – create an event data store, write Lake queries for IAM changes & S3 policy updates, and set up SNS alerts.		Lab manual Google drive link
30 & 31		Experiment 9: Migrate an EC2-based Web Application from Source AWS Account to Target AWS Account.		Lab manual Google drive link
32	4	Migration Fundamentals: Why Migrate to Cloud? Migration Challenges and Considerations	Lecture on benefits like cost savings and scalability; Group discussion on real-world challenges (e.g., data loss, downtime)	Text Book 3: AWS Well-Architected Framework Migration Guide aws.amazon
33		Migration Planning: Assessing Existing Infrastructure, Cloud Readiness and Choosing a Model	Lecture and Case study analysis	Reference Book 3: Ch. 3 Syllabus.pdf; Google Cloud Adoption Framework
34		Migration Strategies (6 Rs): Rehost, Replatform, Repurchase, Refactor, Retain, Retire	Overview lecture	Text Book 3: Ch. 4 AWS 6 Rs Blog 6 Strategies for Migrating Applications to the Cloud AWS Executive in Residence Blog
35		Migration Execution: Migration Phases: Discover → Plan → Migrate → Operate	Phased flowchart walkthrough	Reference Book 3: Ch. 5 AWS Migration Phases Documentation Iterating the 6 Rs migration strategy selection - AWS Prescriptive Guidance
36		Migration Tools (AWS Migrate, Azure Migrate, GCP Migrate)	Live demos of AWS/Azure/GCP Migrate tools Comparison discussion	Text Book 3: Ch. 6 Azure Migrate Overview
37		Post-Migration Performance Tuning	Best practices lecture	Reference Book 1: GCP Migration Center Guide 10 Google Cloud Migration Tools and Services in 2026

38 & 39		Experiment 5: AWS Application Migration Service (MGN) – Migrating an existing server or application to AWS with minimal changes using a rehosting strategy.	Lab manual Google drive link
40 & 41		Experiment 6: AWS Database Migration Service (DMS) – Migrating relational databases to AWS with minimal downtime.	Lab manual Google drive link
42	5	Security Challenges During Migration: Data Exposure & Transit Risk, Legacy System Threats	Lecture on risks, Threat modeling
43		Secure Migration Practices: Data Encryption (At Rest & In Transit), VPN, Dedicated Connections, IAM Hardening	Lecture
44		Compliance and Governance: Real-Time Logging, Audit Trails, Monitoring, Cloud-Native Security Integration	Logging setup walkthrough (CloudTrail/Stackdriver, Compliance checklist activity
45		DevSecOps & Continuous Security: Integrating Security into CI/CD Pipelines	CI/CD pipeline demo with security scans
46		Secure Multi-cloud and Hybrid Environments	Architecture diagramming for hybrid setups (25 min); Tool compatibility discussion (AWS/GCP/Azure, 20 min); Risk mitigation
47		Case Studies: Real-Life Secure Cloud Migrations, Industry Best Practices	Present 2-3 cases (Agave Pay, RupeeRedee); Group analysis and best practice extraction
48 & 49		Experiment 7: Migration of Object Storage Data Between Amazon S3 Buckets Using AWS DataSync.	

50 & 51		Experiment 10: GuardDuty detects the behavior and automatically isolates the instance from the network.	
52 & 53		Labs Revision & Discussion	
54 & 55		Labs Revision & Discussion	
56	CIE-3 Project Presentations	Student Project and Case study presentations	
57		Student Project and Case study presentations	
58		Student Project and Case study presentations	
59		Student Project and Case study presentations	
60		Overall course summary and CIE finalization	

Assessment and Evaluation

Continuous Internal Evaluation (CIE) – 70%	
Components of CIE	Weightage of each component
CIE-1: Graded Component 1 (Quiz) Scenario based MCQs	20 marks out of 70
CIE-2: Theory exam	25 marks out of 70
CIE-3: I. Continuous Lab Evaluation II. Project – Entire architecture review, Explanation of architecture diagram and services used, Implementation and Result analysis	25 marks out of 70 10 marks out of 25 15 marks out of 25

Semester End Exams (SEE) – 30%	
Mode of Exam	Theory
Weightage of exam	30%

--	--

Course Outcomes- Programme Outcomes Matrix

B.Tech (Hons)

Course Outcomes		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	P O1 2
C01	Understand fundamental cloud security concepts, including service models (IaaS, PaaS, SaaS), deployment models, security architecture, and the shared responsibility model.	3	3	3	1	3	2	-	-	1	1	-	2
C02	Apply identity and access management (IAM) mechanisms, data protection strategies (e.g., encryption), and compliance enforcement techniques in cloud environments.	3	3	3	1	3	2	-	-	1	1	-	2
C03	Design and implement network security configurations, conduct vulnerability assessments, and develop incident response	3	3	3	1	3	2	-	-	1	1	-	2

	procedures for cloud-based systems.												
C04	CO4: Secure Cloud Migration Planning Plan and execute cloud migrations using industry best practices and tools, integrating security across the lifecycle with DevSecOps, governance, monitoring, performance, cost, and reliability considerations.	3	3	3	1	3	2	-	-	1	1	-	2

Rubrics for evaluation of CIE (separate for each CIE component)

Assessment Component	Type of Component	Rubrics for Assessment
1	Quiz	Quiz may be conducted for 20. Each right answer carries two marks each. No negative marking for wrong answers.
2	Theory exam	Test will be conducted for 25 marks and will be evaluated according to Scheme of Evaluation prepared by team of course faculty
4	Practical Component (Lab & Project)	Out of 25 (100 marks scaled down to 25) P.S: Rubrics discussed in Table 1

Table 1: Lab evaluation and Project Presentation Rubrics (Max: 25 M)

Sl. No	Evaluation criteria	Type	Excellent / Good	Average	Needs Improvement
1	Presentation Skills Communication Clarity Confidence (5 marks)	Observations	Engaging, clear, and well-structured presentation that effectively communicates key points. (4 M)	Presentation is somewhat unclear or disorganized, with noticeable room for improvement. (2-3 M)	Presentation is missing or significantly incomplete. (0-1 M)
2	Technical Understanding (7 marks)	Explanation of services	Demonstrates a deep understanding of AWS services used in the project. Able to explain complex concepts clearly. (6-7 M)	Shows good understanding of AWS services, with minor gaps in technical details. (3-5 M)	Lacks understanding of AWS services and cannot effectively explain the solution. (0-2 M)
3	Peer Evaluation (3 marks)	Observation and participation	Asks relevant questions during other teams' presentations (3 M)	Was rarely participating in other teams' presentations. (1 M)	Did not participate in the discussions (0 M)
4	Report (5 marks)	Document submission	Report is well-organized, with clear headings, subheadings, and logical flow of sections. (5 M)	Report is mostly organized, but some sections may be out of order or unclear. (3 M)	Report is well-organized, with clear headings, subheadings, and logical flow of sections. (0-1 M)
5	Subject Knowledge (5 marks)	Viva Voce	Demonstrates deep understanding, answering questions with clarity and depth.	Provides satisfactory answers but lacks depth or clarity.	Unable to answer questions effectively. (0-1 M)

			(4-5 M)	(2-3 M)	
--	--	--	---------	---------	--

